

Exact Resource Frontiers for Residual-Faithful Deterministic Pushdown Automata

State–Stack Tradeoffs for Rational Binary Counting Quotient Graphs

Alp Eren Bütün
Independent Researcher
github.com/alperenbutun

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Abstract

For coprime positive integers p, q , consider the rational binary counting language

$$L_{p,q} = \{w \in \{0,1\}^* : qN_1(w) = pN_0(w)\}$$

and its residual translation graph

$$r \xrightarrow{0} r - p, \quad r \xrightarrow{1} r + q.$$

We study exact descriptive frontiers for deterministic pushdown representations that are required to realize this residual dynamics faithfully, rather than the unrestricted recognition complexity of $L_{p,q}$. In the weakest model, a permanent-bottom DPDA configuration is assigned injectively to every residual, no ε -normalization is permitted, and each residual edge must be one ordinary input transition. If g denotes the number of work-stack symbols, we prove the exact frontier

$$S_{\text{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

Thus one work symbol forces $p + q$ states, whereas two work symbols suffice with only $\max(p, q)$ states. The lower bound uses tail escape, last-height stabilization, synchronization of residue classes, and a cross-sign argument that becomes sharp when the physical stack is unary.

We then show that the optimal three-symbol point survives a push-at-most-two convention: with a right endmarker there is a deterministic implementation with $\max(p, q)$ states and stack alphabet $\{0, 1, Z_0\}$, while ordinary final-state acceptance without an endmarker requires exactly one additional control state in the corresponding raw-conjugacy model. Long unary writes are compiled into forced suffix-sharing chains; exactly $|p - q|$ transient roles are needed and they fit into state/top cells unused by stable semantics.

Two more structured representation budgets are solved exactly. A phase-local B -coordinate architecture with one-symbol visibility and C work symbols has

$$Q_{\min}^{\text{phase}}(B, C) = B \left\lceil \frac{p/\gcd(B, p) + q/\gcd(B, q)}{C} \right\rceil.$$

Finally, in a direct lossless unary-tail factorization model with push-at-most-two, the stable semantic-cell count is $2p + 2q - 1$ and the exact transition count is

$$4(p + q) - 1 + |p - q|.$$

The results are deliberately scoped to the stated residual-faithful models; no unrestricted DPDA state-complexity claim for $L_{p,q}$ is made.

Keywords: deterministic pushdown automata; descriptive complexity; stack alphabet; residual graph; state–stack tradeoff; counting languages; pushdown configurations.

1 Introduction

The language $L_{p,q}$ is easy to describe arithmetically: after a prefix w , the signed residual

$$\rho(w) = qN_1(w) - pN_0(w)$$

changes by $-p$ on input 0 and by $+q$ on input 1. Deterministic counter-like recognition of such weighted equalities lies inside classical one-counter territory [1]. The question here is different. We ask how much finite control and how many stack symbols are required when a pushdown representation must preserve the residual dynamics itself.

This distinction matters. A recognizer may encode the same language by an arbitrary internal mechanism. By contrast, a residual-faithful representation is constrained semantically: configurations stand for residuals, and input transitions must implement the two translations of the quotient graph. A companion manuscript on the same counting family already establishes the exact one-work-symbol raw endpoint $|Q| = p + q$, gives an explicit three-symbol raw realization with $\max(p, q)$ states, and proves the phase-capacity lower bound for the remainder-separated architecture [10]. The new contribution here is to close those endpoints into full frontiers: we prove the representation-independent $g \geq 2$ raw lower bound, give the matching phase-local upper construction, preserve the optimal three-symbol point under push-at-most-two, and derive exact direct-factorization transition costs. Thus the present paper is a frontier-closing sequel rather than a re-presentation of the companion results.

The first and strongest level imposes almost no stack-shape hypothesis. A permanent bottom marker is fixed, the residual-to-configuration map is injective, there are no ε -normalization moves, and each quotient edge is one actual input transition. We call this *raw residual conjugacy*. The headline theorem is the complete state/work-alphabet frontier

$$S_{\text{raw}}(g) = \infty, \quad p + q, \quad \max(p, q)$$

for $g = 0, 1, \geq 2$, respectively. The proof is not a homogeneous-tail argument. It uses only injectivity, finiteness, the fact that one PDA transition can remove at most the visible top work symbol, and arithmetic synchronization of residue classes.

The second level asks whether the optimal three-symbol point survives a standard bounded-push convention. It does. For $p \geq q$, the only potentially long writes are unary 1-writes on the negative side. Their remaining pushes can be shared by suffix chains. The exact number of transient suffix roles is $p - q$, which is also the number of stable state/top-1 cells left unused by negative semantics. A canonical overlay therefore preserves the $\max(p, q)$ state count.

The third and fourth results intentionally impose additional semantic architecture. In a phase-local B -coordinate model, a finite remainder and a locally visible signed phase must be represented explicitly; the exact state cost becomes a packing formula. In a direct lossless unary-tail factorization model, the semantic role graph itself has a fixed number of stable state/top cells, and bounded push forces an exact number of persistent ε -push rules. These results should be read as exact descriptive theorems inside specified models, not as competing global minima.

Contributions. The paper proves the following.

- (i) The exact unrestricted raw-conjugacy frontier with permanent bottom marker:

$$S_{\text{raw}}(0) = \infty, \quad S_{\text{raw}}(1) = p + q, \quad S_{\text{raw}}(g) = \max(p, q) \quad (g \geq 2).$$

- (ii) The exact $+1$ control-state cost for ordinary final-state acceptance without a right end-marker in the same raw model.

- (iii) A three-symbol, $\max(p, q)$ -state right-endmarker realization with every replacement word of length at most two, together with the symmetric $\max(p, q) + 1$ -state no-endmarker version.
- (iv) The exact one-symbol-visible phase-local frontier

$$Q_{\min}^{\text{phase}}(B, C) = B \lceil H(B)/C \rceil, \quad H(B) = \frac{p}{\gcd(B, p)} + \frac{q}{\gcd(B, q)}.$$

- (v) In a direct lossless unary-tail factorization model, the exact stable semantic-cell count $2p + 2q - 1$, exact forced- ε count $|p - q|$, and exact total transition count $4(p + q) - 1 + |p - q|$ under push-at-most-two.

Scope. State/stack-symbol tradeoffs and stack-alphabet hierarchies are established themes in pushdown descriptonal complexity [2, 3, 4, 5, 6]. The contribution claimed here is the exact family-specific frontier for the explicitly defined residual-conjugacy model and the exact nested frontiers that follow after stronger semantic requirements are imposed. In particular, we do *not* claim the unrestricted DPDA state complexity of $L_{p,q}$.

2 Residual graph and implementation models

Throughout, $p, q \geq 1$ are coprime and

$$M = \max(p, q), \quad m = \min(p, q).$$

Let $R_{p,q}$ denote the labeled graph on $\mathbb{Z}$ with edges

$$r \xrightarrow{0} r - p, \quad r \xrightarrow{1} r + q.$$

Because $\gcd(p, q) = 1$, every integer occurs as a residual of some binary word.

2.1 Raw residual conjugacy

Definition 2.1 (Model R). Let $\Gamma = \Delta \cup \{Z_0\}$, where Z_0 is a permanent bottom marker and $g = |\Delta|$. A *raw residual conjugacy* is an injective map

$$\eta : \mathbb{Z} \longrightarrow Q \times \Delta^* Z_0$$

such that no ε -transition is used on encoded residual configurations and

$$\delta(\eta(r), 0) = \eta(r - p), \quad \delta(\eta(r), 1) = \eta(r + q)$$

for every $r \in \mathbb{Z}$.

No homogeneous-tail or sign-separated hypothesis is imposed. Nor do we assume fixed depth, bounded replacement, or a canonical coordinate form. The transition relation may replace the visible top work symbol by any fixed finite word allowed by the machine.

Let $S_{\text{raw}}(g)$ be the least number of control states in Model R with g work symbols, with ∞ if no finite-state realization exists.

2.2 Normalized macro-step implementations

For bounded-push constructions we also use a weaker edge requirement: one input-consuming move may be followed by a uniquely forced terminating ε -burst. Stable configurations are still indexed by residuals, but the quotient edge is represented by a deterministic macro-step rather than literally by one transition. This distinction is made explicit whenever used.

2.3 Phase-local B -coordinate architecture

Fix $B \geq M$ and write every residual uniquely as

$$r = s + Bz, \quad 0 \leq s < B, \quad z \in \mathbb{Z}.$$

Define

$$h_+(B) = \frac{p}{\gcd(B, p)}, \quad h_-(B) = \frac{q}{\gcd(B, q)}, \quad H(B) = h_+(B) + h_-(B).$$

The positive translation orbit of a remainder under repeated subtraction by p has period $h_+(B)$ modulo B ; the negative translation orbit under repeated addition by q has period $h_-(B)$. In the *phase-local* architecture, each remainder s carries $H(B)$ signed quotient-phase roles and the currently active phase must be determined by the control state together with the single visible work symbol. With C available work symbols, one control state supplies at most C such state/top cells for a fixed remainder. This is a semantic representation requirement, not a property demanded of arbitrary recognizing DPDAs.

2.4 Direct lossless unary-tail factorization

The final model fixes a particular lossless factorization of the residual representation into a bounded semantic frontier and an unbounded homogeneous tail. Stable roles fall into three disjoint categories:

- (a) $p + q - 1$ boundary/bottom roles;
- (b) p deep positive roles whose visible tail symbol is 0;
- (c) q deep negative roles whose visible tail symbol is 1.

The model requires these semantic roles to remain distinguishable as state/top cells. Transient stack symbols and deterministic ε -normalization are otherwise unrestricted, except that replacement length is at most two in the transition-complexity theorem.

3 Boundary-compatible upper constructions

We first give the upper points needed for the exact frontier. The two-work-symbol construction is also the semantic base for the bounded-push compiler.

3.1 Two work symbols and M states

Assume $p \geq q$; the case $q > p$ follows by exchanging 0 with 1, p with q , and r with $-r$. Let

$$Q = \{R_0, \dots, R_{p-1}\}, \quad \Gamma = \{0, 1, Z_0\}.$$

For $r \geq 0$, write $r = pk + s$ with $0 \leq s < p$ and set

$$\zeta(r) = (R_s, 0^k Z_0).$$

For $r < 0$, write uniquely

$$r = u - qh, \quad 0 \leq u < q, \quad h \geq 1,$$

and set

$$\zeta(r) = (R_u, 1^h Z_0).$$

For $0 \leq s < p$, define

$$s^+ = (s + q) \bmod p, \quad c_s = \frac{s + q - s^+}{p} \in \{0, 1\},$$

and

$$u_s = (s - p) \bmod q, \quad h_s = \frac{u_s + p - s}{q} \geq 1.$$

For $0 \leq u < q$, define

$$u^- = (u - p) \bmod q, \quad d_u = \frac{u^- - u + p}{q} \geq 1.$$

The raw transition schema is

$$\begin{aligned} \delta(R_s, 0, 0) &= (R_s, \varepsilon), & \delta(R_s, 1, 0) &= (R_{s^+}, 0^{c_s+1}), \\ \delta(R_s, 1, Z_0) &= (R_{s^+}, 0^{c_s} Z_0), & \delta(R_s, 0, Z_0) &= (R_{u_s}, 1^{h_s} Z_0), \\ \delta(R_u, 1, 1) &= (R_u, \varepsilon), & \delta(R_u, 0, 1) &= (R_{u^-}, 1^{d_u+1}). \end{aligned}$$

A direct substitution verifies that every transition realizes exactly one residual edge.

Proposition 3.1. *For every coprime p, q , Model R has a realization with M states and two work symbols.*

Proof. The displayed construction handles $p \geq q$. Its stable encodings are injective because sign is visible in the homogeneous tail, and within each sign the state stores the Euclidean remainder while the height stores the quotient. The six transition families above preserve the defining residual equations. The involution $0 \leftrightarrow 1$, $p \leftrightarrow q$, $r \leftrightarrow -r$ gives the symmetric construction when $q > p$. $\square$

3.2 One work symbol and $p + q$ states

Let $\Gamma = \{A, Z_0\}$. Use disjoint state families

$$P_0, \dots, P_{p-1}, \quad N_0, \dots, N_{q-1}.$$

For $r \geq 0$, write $r = pk + s$ and encode

$$\xi(r) = (P_s, A^k Z_0).$$

For $r < 0$, write $r = u - qh$ with $0 \leq u < q$, $h \geq 1$, and encode

$$\xi(r) = (N_u, A^{h-1} Z_0).$$

The offset $h - 1$ moves the negative sign-changing boundary to Z_0 . Each residual update can then be implemented in one input transition by Euclidean division of the new remainder and the corresponding fixed unary replacement. We omit the routine table because the lower bound below is the substantive part; the construction uses exactly $p + q$ states.

Proposition 3.2. *Model R with one work symbol has a realization with $p + q$ states.*

4 Exact raw-conjugacy frontier

We now prove the headline lower bound without assuming a regular stack shape.

For a configuration $(q, \alpha Z_0)$, let $h(q, \alpha Z_0) = |\alpha|$ be its work-stack height.

4.1 Tail escape and last-height stabilization

Lemma 4.1 (Tail escape). *Fix a positive residue class $s \bmod p$. Then*

$$h(\eta(s + pk)) \longrightarrow \infty \quad (k \rightarrow \infty).$$

The analogous statement holds on every negative residue class modulo q .

Proof. If infinitely many encoded residuals on the chain had height at most H , then only finitely many physical configurations could occur: Q and Δ are finite and there are only finitely many stack words of height at most H . This contradicts injectivity of η . $\square$

Fix a positive residue class and write

$$c_k = \eta(s + pk), \quad h_k = h(c_k).$$

Since $c_{k+1} \xrightarrow{0} c_k$, one ordinary PDA transition removes at most the visible top work symbol. Hence

$$h_{k+1} \leq h_k + 1.$$

Lemma 4.2 (Last-height crossing). *For every sufficiently large integer H , the sequence (h_k) has a last occurrence of height H . If it occurs at index k , then $h_{k+1} = H + 1$ and the transition $c_{k+1} \xrightarrow{0} c_k$ is a pure one-symbol pop. All later configurations on the chain have height $> H$ and share a common bottom- H work-stack suffix.*

Proof. By theorem 4.1, $h_k \rightarrow \infty$. The one-sided bound $h_{k+1} \leq h_k + 1$ implies that every sufficiently large height is attained: a jump from below H to above H cannot skip H in the forward indexed direction. Let k be the final index with $h_k = H$. If $h_{k+1} \leq H$, then either $h_{k+1} = H$, contradicting finality, or a later rise to heights above H would have to pass through H again. Thus $h_{k+1} = H + 1$.

The transition from height $H + 1$ to height H can only delete the visible top work symbol; it therefore leaves the remaining H symbols unchanged. After the last height- H occurrence every later configuration has height $> H$, and each transition among such configurations modifies only the top cell. Consequently the bottom H work symbols remain fixed forever along the tail. $\square$

4.2 Synchronization across residue classes

For each positive residue class $s \in \mathbb{Z}_p$, choose the last-height- H configuration and let $\beta_s \in \Delta^H$ be its work-stack word. By theorem 4.2, β_s is also the stabilized bottom- H suffix of the later tail.

Lemma 4.3 (Positive synchronization). *For sufficiently large H , all positive residue classes have the same last-height- H stack word. Consequently at least p distinct control states occur among those configurations.*

Proof. For sufficiently deep positive residuals, input 1 maps the tail of residue class s into the tail of residue class

$$s + q \pmod{p}.$$

Choose the depth so that both source and target lie above height H . A one-step transition cannot touch the stabilized bottom- H suffix, so $\beta_s = \beta_{s+q}$. Since $\gcd(p, q) = 1$, addition by q is a single p -cycle on $\mathbb{Z}_p$. Hence all β_s coincide.

At the selected last-height cross-sections the complete physical stack is exactly this common word followed by Z_0 . The encoded residuals are distinct, so injectivity forces their control states to be pairwise distinct. $\square$

The symmetric argument yields:

Lemma 4.4 (Negative synchronization). *At least q distinct control states are required on the negative tail.*

Together, theorems 4.3 and 4.4 give $|Q| \geq \max(p, q)$ for every nonempty work alphabet.

4.3 Unary cross-sign sharpening

When $g = 1$, write $\Delta = \{A\}$. Every height- H work stack is the same word A^H . We may choose the synchronized positive and negative cross-sections at one common sufficiently large height.

Lemma 4.5 (Cross-sign unary separation). *If $g = 1$, the p synchronized positive states and the q synchronized negative states are pairwise disjoint. Hence $|Q| \geq p + q$.*

Proof. All selected configurations have the identical physical stack $A^H Z_0$. They encode $p + q$ distinct residuals, chosen far enough from zero that the positive and negative sets are disjoint. Injectivity of η therefore forces all $p + q$ control states to be distinct. $\square$

Theorem 4.6 (Exact raw frontier). *For every coprime $p, q \geq 1$,*

$$S_{\text{raw}}(g) = \begin{cases} \infty, & g = 0, \\ p + q, & g = 1, \\ \max(p, q), & g \geq 2. \end{cases}$$

Proof. With $g = 0$, the permanent-bottom stack admits only finitely many configurations, so no injective encoding of $\mathbb{Z}$ exists. For $g \geq 1$, theorems 4.3 and 4.4 give the lower bound $|Q| \geq \max(p, q)$; when $g = 1$, theorem 4.5 strengthens it to $p + q$. The matching upper bounds are theorems 3.1 and 3.2; extra work symbols may be left unused. $\square$

Remark 4.7. The proof never assumes that encoded tails are homogeneous, sign-separated, or generated by a fixed coordinate formula. Those properties appear only in the matching upper constructions.

4.4 No-endmarker acceptance

For this corollary we take the machine's initial configuration to be the encoding of residual zero and require full recognition of $L_{p,q}$ after complete input consumption. Under ordinary final-state acceptance, a state used by any nonzero stable residual cannot be final, because an input may end at that residual. The synchronized lower-bound states therefore cannot double as the unique accepting zero role.

Corollary 4.8 (Exact no-endmarker raw cost). *For standard final-state acceptance without a right endmarker,*

$$S_{\text{raw}}^{\text{noend}}(g) = \begin{cases} \infty, & g = 0, \\ p + q + 1, & g = 1, \\ \max(p, q) + 1, & g \geq 2. \end{cases}$$

Proof. The lower-bound states in theorem 4.6 all occur on nonzero residual configurations and hence must be nonfinal. Residual zero requires a separate final state. Conversely, add one final zero state to the matching raw constructions and redirect exactly the transitions whose semantic target is zero. $\square$

5 Bounded-push preservation at the optimal three-symbol point

The raw construction of theorem 3.1 may use long homogeneous replacement words. We now show that the state/alphabet optimum survives the convention $|\alpha| \leq 2$ for every replacement word α , at the price of forced ε -normalization.

Assume $p \geq q$. The positive-side replacement lengths are already at most two because $c_s \in \{0, 1\}$. Only unary 1-pushes on the negative side and at nonnegative-to-negative crossings can be long.

For each target $v \in \mathbb{Z}_q$, let

$$\text{pre}(v) = (v + p) \bmod q, \quad L_v = d_{\text{pre}(v)} - 1.$$

Since addition by p permutes $\mathbb{Z}_q$,

$$\sum_{v=0}^{q-1} L_v = \sum_{u=0}^{q-1} (d_u - 1) = p - q.$$

Introduce an abstract suffix role $H_{v,j}$, $1 \leq j \leq L_v$, meaning “add exactly j more unary 1’s, then enter R_v .” Its forced rules are

$$\begin{aligned} \delta(H_{v,1}, \varepsilon, 1) &= (R_v, 11), \\ \delta(H_{v,j}, \varepsilon, 1) &= (H_{v,j-1}, 11), \quad j > 1. \end{aligned}$$

A deep negative rule requiring $1 \mapsto 1^{d_u+1}$ consumes the input by $1 \mapsto 11$ and, when $d_u > 1$, enters H_{u^-, d_u-1} .

The key point is that these are roles, not new state names. Stable negative configurations use top 1 only in states $R_0, \dots, R_{q-1}$. Therefore the cells

$$(R_i, 1), \quad q \leq i < p,$$

are unused. There are exactly $p - q$ of them, matching the total number of suffix roles. Fix any bijection between roles and these cells.

Boundary writes reuse the same chains. For a boundary residual s requiring h_s unary 1’s, the consuming move writes $1Z_0$ and, if $h_s > 1$, enters the role with $h_s - 1$ additions remaining. This role exists because, with $u_s = (s - p) \bmod q$,

$$d_{\text{pre}(u_s)} = h_s + \left\lfloor \frac{s}{q} \right\rfloor,$$

so

$$h_s - 1 \leq L_{u_s}.$$

Theorem 5.1 (State-preserving bounded-push compiler). *For every coprime p, q , there is a deterministic right-endmarker DPDA whose normalized stable configurations realize the residual graph and satisfy*

$$\boxed{|Q| = \max(p, q), \quad |\Gamma| = 3, \quad \max |\alpha| \leq 2.}$$

The only ε -moves are uniquely forced suffix-chain moves.

Proof. The construction above proves the case $p \geq q$. Every long negative write is split into one consuming push and a strictly descending chain of suffix roles. The role count is exactly $p - q$ and all roles fit into unused state/top-1 cells, so no state or stack symbol is added. Because the base machine has no consuming transition on those cells, overlay introduces no ε /input conflict. Boundary crossings enter suffixes of the same chains. Every chain terminates and its macro-result equals the corresponding raw transition. The case $q > p$ is symmetric. $\square$

Corollary 5.2 (No-endmarker bounded push). *Under standard final-state acceptance without a right endmarker, there is a normalized residual-faithful DPDA with*

$$\boxed{|Q| = \max(p, q) + 1, \quad |\Gamma| = 3, \quad \max |\alpha| \leq 2.}$$

Proof. Use one fresh final state Z for residual zero. Exact-multiple pops remain in nonfinal R_0 ; only after Z_0 is exposed does the forced rule

$$\delta(R_0, \varepsilon, Z_0) = (Z, Z_0)$$

enter the final state. Thus no input ending before the bottom marker is exposed is accepted prematurely. Zero-source binary moves are defined on (Z, Z_0) and use the same boundary formulas and shared suffix chains. $\square$

6 Exact phase-local frontier

We next solve the explicitly more local B -coordinate architecture from section 6. The point is not that arbitrary DPDAs must expose these phases, but that if this semantic visibility is required then the state/alphabet tradeoff is exact.

Lemma 6.1 (Phase capacity). *Fix a remainder $s \in \{0, \dots, B-1\}$. In a one-symbol-visible phase-local realization, the $H(B)$ signed phase roles associated with s require at least*

$$\left\lceil \frac{H(B)}{C} \right\rceil$$

control states.

Proof. By definition of the architecture, different signed phases at the same remainder must be distinguishable from the pair consisting of the control state and the visible work symbol. A fixed control state provides at most C such work-symbol cells. Packing $H(B)$ distinct roles therefore needs at least the stated number of states. $\square$

Remainders are also semantically distinct, so their state blocks cannot be identified in this architecture.

Theorem 6.2 (Exact normalized phase-local frontier). *For $B \geq \max(p, q)$ and $C \geq 1$,*

$$\boxed{Q_{\min}^{\text{phase}}(B, C) = B \left\lceil \frac{H(B)}{C} \right\rceil, \quad H(B) = \frac{p}{\gcd(B, p)} + \frac{q}{\gcd(B, q)}.$$

For $C \geq 2$ the matching construction can be made ε -free and one-input-step raw. For $C = 1$ the same normalized state bound is attained with one forced bottom-cell ε -normalization at the negative zero crossing.

Proof. The lower bound follows by summing theorem 6.1 over the B remainders. For the upper bound, write every residual uniquely as $r = s + Bz$ and define a universal signed quotient phase from z : on the positive side record $z \bmod h_+(B)$, and on the negative side record $(-z - 1) \bmod h_-(B)$. For each fixed remainder s , the canonical signed phase is a bijective function of this universal phase and s . Inject the resulting $H(B)$ phase labels into

$$\{0, \dots, c-1\} \times \{0, \dots, C-1\}, \quad c = \lceil H(B)/C \rceil,$$

and use states $q_{s,j}$ together with work symbols $K_0, \dots, K_{C-1}$. The stack stores one coded symbol per nonzero quotient level, so its height is $|z|$ and the state/top pair determines the signed phase.

For an input residual increment $d \in \{q, -p\}$, let

$$s' = (s + d) \bmod B, \quad a = \frac{s + d - s'}{B}.$$

Because $B \geq \max(p, q)$, one has $a \in \{-1, 0, 1\}$ and $z' = z + a$. Thus each edge is implemented by one top operation: if $a = 0$ the stack is unchanged; moving one level away from zero pushes the code symbol for the new phase; moving one level toward zero pops the visible code symbol. Control changes from the row for $(s, \phi(z))$ to the row for $(s', \phi(z'))$. This realizes every residual edge while using exactly Bc states.

For $C \geq 2$, the boundary phase codes can be chosen so that the sign-changing cases are distinguished in the visible state/top cell, giving an ε -free one-input-step realization. When $C = 1$, the recurring negative phase at the final level and at deeper levels cannot be distinguished before a pop. The pop always enters the deeper negative target row; if it was the boundary pop, Z_0 is exposed and that otherwise unused bottom cell performs one forced ε -move to the zero-depth row. No new state is introduced, so the same normalized state bound is attained. $\square$

Corollary 6.3 (A superlinear exact family). *Let*

$$p = k^2, \quad q = k^2 - 1, \quad B = k(k + 1).$$

Then $H(B) = 2k - 1$ and

$$Q_{\min}^{\text{phase}}(B, C) = k(k + 1) \left\lceil \frac{2k - 1}{C} \right\rceil.$$

For every fixed C , this is $\Theta(k^3) = \Theta(M^{3/2})$.

7 Direct-factorization description complexity

The direct lossless unary-tail model fixes more semantic structure than either Model R or the phase-local architecture. Its value is that the number of stable semantic cells and the transition budget can both be counted exactly.

7.1 Stable semantic cells

Theorem 7.1 (Exact semantic-cell count). *Every exact implementation of the direct lossless unary-tail semantic graph requires, and the canonical semantic core attains,*

$$C_{\text{sem}, \min} = 2p + 2q - 1$$

reachable stable state/top cells.

Proof. The model has three disjoint stable role sets: $p + q - 1$ boundary/bottom roles, p deep positive top-0 roles, and q deep negative top-1 roles. Losslessness requires distinct semantic roles to be represented by distinct state/top cells. Hence at least their sum is necessary, and the direct factorization uses exactly one cell per role. $\square$

Each stable cell has two binary input actions. Therefore the exact binary-rule baseline is

$$2(2p + 2q - 1) = 4(p + q) - 2.$$

The right-endmarker convention adds one accepting rule at residual zero.

7.2 Persistent pushes

Assume $p > q$; the other case is symmetric. For a deep negative remainder t , the bounded-push normalization associated with an input-0 update must contribute net height

$$D_t = \left\lfloor \frac{p+t}{q} \right\rfloor - 1.$$

Lemma 7.2 (Persistent-push lemma). *Any terminating, bottom-avoiding deterministic ε -computation with net stack-height increase D contains at least D distinct transition rules whose pushed symbol survives to the end of the computation.*

Proof. Choose the D net new stack levels that remain present at termination and, for each, the transition that first creates that level. If the same state/top transition rule created two such persistent levels, the computation segment between those two occurrences would be a context-independent positive-drift ε -macro from the same state/top mode back to itself while never exposing the lower context. Determinism would allow that macro to repeat indefinitely, contradicting termination. $\square$

Lemma 7.3 (Cross-remainder separation). *Persistent ε -push rules used for two distinct source remainders cannot coincide.*

Proof. Suppose one persistent rule were shared. Immediately after that rule, both normalization bursts have the same control/top prefix, and the newly persistent level prevents the lower contexts from being exposed. The subsequent deterministic ε -behavior is therefore identical. In the direct lossless factorization, however, the target remainder modulo q is encoded by the stable top-1 semantic cell, not by absolute lower-stack depth. The map

$$t \mapsto t + p \pmod{q}$$

is a permutation, so different source remainders require different target remainder cells. This contradicts identical deterministic continuation. $\square$

The arithmetic identity

$$\sum_{t=0}^{q-1} D_t = p - q$$

then gives the exact forced normalization cost.

Theorem 7.4 (Exact ε -transition count). *In the direct lossless unary-tail factorization model with push-at-most-two and a right endmarker,*

$$\boxed{|\delta_\varepsilon|_{\min} = |p - q|}.$$

Proof. For $p > q$, theorems 7.2 and 7.3 give at least $\sum_t D_t = p - q$ pairwise distinct persistent ε rules. The canonical suffix-chain implementation attains exactly that number. Symmetry gives $q - p$ when $q > p$. $\square$

Theorem 7.5 (Exact total transition count). *Under the same model,*

$$\boxed{|\delta|_{\min} = 4(p + q) - 1 + |p - q|}.$$

Proof. The semantic graph forces $4(p + q) - 2$ binary input rules by theorem 7.1, the right-endmarker convention forces one accepting rule, and theorem 7.4 forces $|p - q|$ normalization rules. The direct construction realizes all three baselines simultaneously. $\square$

8 A top-window implementation lemma

For completeness we record a standard type of finite-control compilation that is useful when comparing semantic visibility budgets.

Proposition 8.1 (Top- λ annotation compiler). *Suppose an abstract pushdown system inspects the top λ work symbols but rewrites only the current top abstract symbol. If the abstract work alphabet has size g , the same one-step dynamics can be implemented with the same control states by annotating physical stack symbols, using at most*

$$\sum_{d=1}^{\lambda} g^d$$

work annotations, plus the bottom marker.

Proof. Annotate each physical symbol by the abstract symbol at that level together with the next at most $\lambda - 1$ abstract symbols below it. The visible physical top then contains the entire abstract window needed to choose the transition. Because only the current abstract top is rewritten, the new annotation can be computed locally from the old annotation and the replacement word. The number of nonempty words of lengths 1 through λ over a g -symbol alphabet gives the stated bound. $\square$

Remark 8.2. The proposition is not stated for an abstract operation that rewrites or pops an entire visible λ -window at once; that stronger claim would require a different compiler.

9 Related work and novelty boundary

Descriptive tradeoffs between control states and stack symbols are classical. Goldstine, Price, and Wotschke studied reductions in the number of PDA stack symbols and the associated state cost [2]; broader surveys treat limited machine resources as a descriptive-complexity problem [3]. Pighizzini’s analysis of unary DPDAs emphasizes state/top-symbol “modes” as a useful local unit [5]. Masopust established stack-alphabet hierarchies for stateless and fixed-state realtime deterministic pushdown automata [4]; later work continues to derive exact simultaneous resource bounds in restricted pushdown models, including input-driven automata [6].

Relative to the companion counting-language manuscript, the one-work-symbol endpoint and the explicit three-symbol upper point are not claimed again as new results. The new raw-conjugacy result is the matching lower bound for every $g \geq 2$, which completes the state/work-alphabet frontier without a prescribed stack shape; the phase-local section likewise adds the matching upper construction to the previously established capacity lower bound.

The present raw-conjugacy theorem is different in scope. It does not ask whether an arbitrary DPDA recognizes $L_{p,q}$ with few resources. It asks for an injective configuration realization of the quotient translation graph with one physical input transition per graph edge and a permanent bottom marker. Within that model, theorem 4.6 is a full frontier and does not prescribe a homogeneous-tail coordinate system.

Configuration graphs and shortest paths in counter/pushdown systems are also established subjects [7, 8]. Here they motivate the semantics but are not claimed as new general machinery. Likewise, using finite control to remember bounded stack information is a standard normalization idea. The bounded-push theorem is therefore presented as a family-specific exact preservation result: the optimal $\max(p, q)$ -state, three-symbol point survives push-at-most-two because the exact transient role count $|p - q|$ matches the number of unused stable state/top cells.

A companion hierarchy manuscript develops a representation-independent recognition lower bound for a different witness family. That result and the present paper should not be conflated: the hierarchy theorem concerns arbitrary equivalent DPDAs, while the strongest theorem here

concerns arbitrary encodings only *within raw residual conjugacy*. The distinction is precisely what allows both results to be exact without making an unsupported global state-minimality claim for $L_{p,q}$ itself.

10 Discussion

The three exact frontiers form a hierarchy of semantic commitments.

Level	Required semantics	Exact resource statement
Raw conjugacy	injective residual configurations; one transition per quotient edge	$S_{\text{raw}}(1) = p + q$, $S_{\text{raw}}(g) = M$ for $g \geq 2$
Phase-local	raw/normalized residual coordinate plus visible signed quotient phase	$B[H(B)/C]$ states
Direct factorization	fixed lossless boundary/tail semantic cells; push-at-most-two	$2p + 2q - 1$ stable cells and $4(p + q) - 1 + p - q $ transitions

The first level is the least architecture-specific and therefore the principal theorem. The second asks how expensive a particular kind of local semantic observability is. The third asks for exact description size after a direct factorization has been fixed. Presenting these as nested scopes prevents a common error in pushdown descriptonal analysis: a lower bound valid for a chosen representation should not be silently promoted to a lower bound for arbitrary recognizing machines.

The most natural remaining question is correspondingly outside the present scope: obtain fixed-total-description lower bounds for arbitrary DPDAs recognizing specific counting languages while allowing the adversary to choose any internal representation. The representation-independent completion-slope method used for a separate hierarchy witness suggests one possible direction, but the arithmetic regularity of $L_{p,q}$ makes such a lower bound substantially more delicate.

11 Conclusion

For the two-generator residual graph of $L_{p,q}$, the cost of faithful pushdown realization has a sharp state/stack-symbol threshold. With a permanent bottom marker and one work symbol, raw residual conjugacy requires exactly $p + q$ states. As soon as a second work symbol is available, the exact state requirement drops to $\max(p, q)$, and this optimum is achieved without imposing any lower-bound assumption on stack shape. The proof is driven by geometric facts about arbitrary encoded tails: heights must escape, last-height cross-sections stabilize a deep suffix, the opposite input synchronizes residue classes, and a unary physical stack prevents positive and negative classes from sharing control states.

The same optimal three-symbol state point survives push-at-most-two normalization because the long unary writes possess an exact suffix-sharing structure. Stronger semantic models then yield their own exact packing and transition frontiers. Together these results separate three notions that are often blurred: language recognition, faithful quotient-graph realization, and faithful realization with additional local semantic structure.

A Explicit bounded-push table for $p \geq q$

This appendix records a canonical state overlay for theorem 5.1. For each target $v \in \mathbb{Z}_q$, let

$$L_v = d_{(v+p) \bmod q} - 1, \quad A_v = \sum_{w=0}^{v-1} L_w,$$

and define, for $1 \leq j \leq L_v$,

$$\chi(v, j) = q + A_v + j - 1.$$

Then χ bijects the abstract suffix roles onto $\{q, \dots, p-1\}$. The role (v, j) is represented by the state/top cell $(R_{\chi(v, j)}, 1)$.

The complete right-endmarker transition families are:

$$\begin{aligned} \delta(R_s, 0, 0) &= (R_s, \varepsilon), \\ \delta(R_s, 1, 0) &= (R_{s+}, 0^{c_s+1}), \\ \delta(R_s, 1, Z_0) &= (R_{s+}, 0^{c_s} Z_0). \end{aligned}$$

For boundary input 0,

$$\delta(R_s, 0, Z_0) = \begin{cases} (R_{u_s}, 1Z_0), & h_s = 1, \\ (R_{\chi(u_s, h_s-1)}, 1Z_0), & h_s > 1. \end{cases}$$

For negative stable cells,

$$\begin{aligned} \delta(R_u, 1, 1) &= (R_u, \varepsilon), \\ \delta(R_u, 0, 1) &= \begin{cases} (R_{u-}, 11), & d_u = 1, \\ (R_{\chi(u-, d_u-1)}, 11), & d_u > 1. \end{cases} \end{aligned}$$

The forced suffix rules are

$$\delta(R_{\chi(v, 1)}, \varepsilon, 1) = (R_v, 11),$$

and, for $j > 1$,

$$\delta(R_{\chi(v, j)}, \varepsilon, 1) = (R_{\chi(v, j-1)}, 11).$$

Finally,

$$\delta(R_0, \neg, Z_0) = (R_0, Z_0).$$

Every replacement has length at most two. The $p - q$ helper cells carry no competing consuming transition, so deterministic ε /input exclusivity is preserved.

B Exact transition count of the explicit compiler

For $p \geq q$, the table above contains $2p$ positive-tail binary rules, $2p$ boundary binary rules, $2q$ negative-tail binary rules, $p - q$ forced ε rules, and one endmarker rule. Hence

$$|\delta| = 5p + q + 1 = 5 \max(p, q) + \min(p, q) + 1.$$

The no-endmarker variant removes the endmarker rule and adds one zero-normalization ε rule, so it has the same count. This is the count of this explicit compiler, not a transition-minimality claim; the exact transition-minimal theorem in theorem 7.5 concerns the different direct-factorization model.

C Executable consistency audit

The bounded-push table was independently compared against the repository implementation of the three-symbol raw schema. For all coprime pairs

$$1 \leq q \leq p \leq 60$$

and residuals $-600 \leq r \leq 600$, the audit checked the raw encoded target, the right-endmarker compiled macro-step, the no-endmarker compiled macro-step, replacement lengths, deterministic ε /input exclusivity, helper-role cardinality, and the explicit transition count. There are 1102 coprime parameter pairs in this range and 2,647,004 residual/input tests in each of the three macro-step categories; all passed. These computations are transcription evidence only and are not proof premises.

Statements and Declarations

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Code and validation artifacts. Validation scripts and checkpoint material are maintained in the author’s public research repositories. The executable checks are supplementary consistency evidence and do not replace the symbolic proofs.

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